

1. What is the probability that a card chosen at random from a standard deck of cards will be either a king or a heart? (Ans: $16/52$ or, $.3077$)

2. A survey by the American Automobile Association (AAA) revealed 60 percent of its members made airline reservations last year. Two members are selected at random. What is the probability both made airline reservations last year? (Ans: $.36$)

3. A golfer has 12 golf shirts in his closet. Suppose 9 of these shirts are white and the others are blue. He gets dressed in the dark, so he just grabs a shirt and puts it on. He plays golf two days in a row and does not do laundry. What is the likelihood both shirts selected are white? (Ans: 0.55)

4. A sample of executives were surveyed about loyalty to their company. One of the questions was, "If you were given an offer by another company equal to or slightly better than your present position, would you remain with the company or take the other position?" The responses of the 200 executives in the survey were cross-classified with their length of service with the company. (See Table 1.)

Table 1: Loyalty of Executives and Length of Service with Company

Loyalty	Less than 1 Year (B ₁)	1-5 Years (B ₂)	6-10 Years (B ₃)	More than 10 Years (B ₄)	Total
Would remain (A ₁)	10	30	55	75	120
Would not remain (A ₂)	25	45	15	105	80
Total	35	75	70	180	200

What is the probability of randomly selecting an executive who is loyal to the company (would remain) and who has more than 10 years of service? (Ans: 0.375)

5. The events A and B are mutually exclusive. Suppose $P(A) = .30$ and $P(B) = .20$. What is the probability of either A or B occurring? What is the probability that neither A nor B will happen? (Ans: $.50$)

6. A study of 200 advertising firms revealed their income after taxes:

Income after Taxes	Number of Firms
Under \$1 million	102
\$1 million to \$20 million	61
\$20 million or more	37

a. What is the probability an advertising firm selected at random has under \$1 million in income after taxes? (Ans: .51)

b. What is the probability an advertising firm selected at random has either an income between \$1 million and \$20 million, or an income of \$20 million or more? What rule of probability was applied? (Ans: .49)

7. Suppose the probability you will get an A in this class is 0.25 and the probability you will get a B is 0.50. What is the probability your grade will be above a C? (Ans: .75)

8. The probabilities of the events A and B are .20 and .30, respectively. The probability that both A and B occur is .15. What is the probability of either A or B occurring? (Ans: .35)

9. Suppose the two events A and B are mutually exclusive. What is the probability of their joint occurrence? (Ans: 0)

10. A survey of grocery stores in the Southeast revealed 40 percent had a pharmacy; 50 percent had a floral shop; and 70 percent had a deli. Suppose 10 percent of the stores have all three departments, 30 percent have both a pharmacy and a deli, 25 percent have both a floral shop and deli, and 20 percent have both a pharmacy and floral shop.

- What is the probability of selecting a store at random and finding it has both a pharmacy and a floral shop?
- What is the probability of selecting a store at random and finding it has both a pharmacy and a deli?
- Are the events "select a store with a deli" and "select a store with a pharmacy" mutually exclusive?
- What is the name given to the event of "selecting a store with a pharmacy, a floral shop, and a deli"?
- What is the probability of selecting a store that does not have all three departments?

(Ans:

- $P(P \text{ and } F) = 0.20$
- $P(P \text{ and } D) = 0.30$
- No
- Joint probability

$P(P \text{ or } D \text{ or } F) = P(P) + P(D) + P(F) - P(P \text{ and } D) - P(P \text{ and } F) - P(F \text{ and } D) + P(P \text{ and } D \text{ and } F)$

$$= .40 + .50 + .70 - .30 = .95$$

11. Suppose $P(A) = .40$ and $P(B | A) = .30$. What is the joint probability of A and B? (Ans: .12)

12. A local bank reports that 80 percent of its customers maintain a checking account, 60 percent have a savings account, and 50 percent have both. If a customer is chosen at random, what is the probability the customer has either a checking or a savings account? What is the probability the customer does not have either a checking or a savings account? (Ans: 0.90, 0.10)

13. Refer to the following table:

	First Event	First Event	First Event	First Event
Second Event	A	A	A	Total
B	2	1	3	6
B	1	2	1	4
Total	3	3	4	10

- a. Determine $P(A)$.
- b. Determine $P(B | A)$.
- c. Determine $P(B \text{ and } A)$.

(Ans:

- a. $P(A) = 3/10 = .30$
- b. $P(B | A) = 1/3 = .33$
- c. $P(B \text{ and } A) = 1/10 = .10$

14. Each salesperson at Puchett, Sheets, and Hogan Insurance Agency is rated either below average, average, or above average with respect to sales ability. Each salesperson is also rated with respect to his or her potential for advancement—either fair, good, or excellent. These traits for the 500 salespeople were cross-classified into the following table:

Sales Ability	Fair	Good	Excellent	Total
Below average	16	12	22	50
Average	45	60	45	150
Above average	93	72	135	300
Total	154	144	202	500

- a. What is this table called?
- b. What is the probability a salesperson selected at random will have above average sales ability and excellent potential for advancement?
- c. Construct a tree diagram showing all the probabilities, conditional probabilities, and joint probabilities.

Answer:

- a. A contingency table
- b. .27, found by $300/500 \times 135/300$

15. The board of directors of a small company consists of five people. Three of those are “strong leaders.” If they buy an idea, the entire board will agree. The other “weak” members have no influence. Three salespeople are scheduled, one after the other, to make sales presentations to a board member of the sales person’s choice. The salespeople are convincing but do not know who the “strong leaders” are. However, they will know who the previous salespeople spoke to. The first salesperson to find a strong leader will win the account. Do the three salespeople have the same chance of winning the account? If not, find their respective probabilities of winning.

Answer:

Probability the first presentation wins = $3/5 = .60$

Probability the second presentation wins = $(2/5)(3/4) = .30$

Probability the third presentation wins = $(2/5)(1/4)(3/3) = .10$